Muhammad **Rifqy**

Cartographer | GIS | Spatial Map Visualizer

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Summary

Results-oriented GIS professional with extensive experience in environmental monitoring and disaster management. Currently serving as a Technical Facilitator for the M4CR Project, overseeing the spatial analysis and mapping of over 5,800 hectares of mangrove rehabilitation areas. Expert in utilizing remote sensing and GIS (ArcGIS, QGIS) to evaluate ecosystem health and guide rehabilitation strategies. Committed to delivering high-quality spatial visualizations that support stakeholder coordination and ecological sustainability.

Skills

Geographical-Related	Map Making, GIS, Land Surveying, Digital Mapping
Geospatial Programming	R (sf, terra, ggplot2, rayshader, etc), Python (Geopandas, Folium, etc), Google Earth Engine (GEE).
Software	ArcGIS, QGIS, SAGA, ILWIS (Basic), Global Mapper, RStudio, Visual Code Studio, Jupyter Notebook
Web-Map	kepler.gl, QGIStoWeb, Mapbox Studio, ArcGIS Online, Open Street Map (Basic)
Languages	Indonesian (Native), English

Education

Universitas Riau
Riau, Indonesia

M.M.B. (Equivalent to MDisMgt in Disaster Management Specialized in Peatland and Coastal Disaster) | GPA 4.00/4.00 — *Aug. 2025 - Now*

- Related Courses: Forest and Peatland Fires, Peatland Ecological Disaster Studies, Disaster Management Information System and GIS

Universitas Gadjah Mada
D.I.Yogyakarta, Indonesia

S.Si. (Equivalent to B.Sc. in Environmental Geography) | CGPA 3.49/4.00 — *Aug. 2017 - Nov. 2022*

- Got a Riau Province Scholarship which is given to promising students who got selected from Penelusuran Bibit Unggul Kemitraan Program
- Related Courses: Geomorphological Survey Mapping, Environmental Geomorphology, Ecology and Environmental Study, Tropical Landscape Ecology, Environmental Degradation, Environmental Management

Experience

Mangrove For Coastal Resilience (M4CR) Project | KEMENHUT-World Bank
Pekanbaru, Riau

Technical Mangrove Rehabilitation Facilitator | Contract — *May 2024-Now*

- Designed and deployed a centralized Shapefile Database System to standardize PRM data management for over 100 community groups, establishing the primary spatial infrastructure for the Project Province Implementation Unit (PPIU) M4CR Riau.
- Managed the spatial analysis for 5,800+ hectares of rehabilitation areas, integrating multi-source data (mangrove health, soil composition, hydrology) to identify optimal planting zones and restoration strategies.
- Engineered a monitoring workflow to track vegetation growth and intervention effectiveness across 1,400 hectares of new planting sites (2025), directly supporting the project's adaptive management plan
- Produced high-precision thematic maps compliant with Ministry standards (P.6/PKTL/SETDIT/2017), serving as the key visual intelligence tools for senior management and stakeholder reporting
- Utilized remote sensing techniques to create detailed maps of mangrove forests, including identifying areas with optimal growth and areas requiring rehabilitation.
- Collaborated with the team in developing a rehabilitation plan, including selecting appropriate mangrove species, determining optimal planting locations, and developing a monitoring strategy.

Extinction Rebellion Chapter Riau
Pekanbaru, Riau

Graphic and Data Visualizer | Volunteer — *August 2023 - July 2025*

- Visualizing the spatial data needed for environmental issues that will be addressed in climate action.
- Analyzing related environmental issues and preparing posters and maps for climate action activities

Laboratory of Environmental Geomorphology and Disaster Mitigation, Universitas Gadjah Mada *D.I.Yogyakarta, Indonesia*

LABORATORY ASSISTANT *2021-2024*

- Helped to develop landslide analysis techniques using the Multinomial Logistic Regression – Regression Kriging method
- Prepared a geocomputational-geostatistics training module focusing on landslide studies using the R programming language
- Checked the results of the practicum reports that have been done.
- Provided practicum material and assessment during the practicum process, both in behavior; knowledge; and activity during the teaching and learning process.
- Courses: GEL0102-Geomorphology, GEL0103-Mineralogy-Petrography, GEL0107-Geomorphological Survey Mapping, GEL0108-Land Survey, Erosion, and Conservation Planning.

Pusat Pengendalian Pembangunan Ekoregion Jawa *D.I.Yogyakarta, Indonesia*

STUDENT INTERN (LAND USE ANALYST) *Mar. 2020 - May. 2020*

- Assisted the process of updating data on the carrying capacity of the environment in West Java Province. Focused on 2 regencies, which are Bogor and Cirebon Regency.
- Delineated for more than 5000 polygons of land use with detail scale (below 1:5000)

Courses

2025	**Bootcamp WebGIS Development**, MAPID	*Bootcamp*
2024	**Bootcamp QGIS: Pemrosesan dan Aplikasi Citra Satelit Resolusi Tinggi**, Geoaccess Indonesia	*Bootcamp*
2024	**Pelatihan GEE Tematik: Pemetaan Lahan Mangrove dan Carbon Stock**, Geoaccess Indonesia	*Course*
2024	**Pelatihan GEE Tematik: Monitoring Perubahan Garis Pantai dan Prediksinya**, Geoaccess Indonesia	*Course*
2022	**Cartography**, ESRI MOOC	*Course*
2022	**Going Places with Spatial Analysis**, ESRI MOOC	*Course*
2022	**Do-It-Yourself Geo Apps**, ESRI MOOC	*Course*
2021	**R Fundamental for Data Science**, DQLab	*Course*
2021	**Python Fundamental for Data Science**, DQLab	*Course*

Recent Projects

Independent Geospatial Research & Visualization (med/rif maps) -

DATA VISUALIZATION & CARTOGRAPHIC DESIGN *Oct. 2023-Now*

- Established a personal geospatial initiative to experiment with advanced cartographic techniques and open-source tools (R, Python, GEE, etc.)
- Portfolio: instagram.com/medrifmaps

Digital Technical Portfolio -

CENTRALIZED PROJECT REPOSITORY *2017-Now*

- A curated repository demonstrating end-to-end GIS workflows, from data acquisition and cleaning (Python/R) to final cartographic production. Includes case studies on land use analysis, disaster mitigation, and web-map development
- Link: mamedrifqy.github.io

Honors & Awards

2022	**Best Map**, Spatialnode Map of The Week	*Global*
2017-2021	**Scholarship Awardee**, The Governance of the Province of Riau's Scholarhip	*Riau, Indonesia*

Publication

Pemanfaatan Inovasi Geospasial Berbasis Sentinel-2 dan UAV untuk Optimalisasi Rehabilitasi Mangrove dalam Kerangka Program Mangroves for Coastal Resilience (M4CR) di Provinsi Riau *ReseachGate*

PREPRINT *Jan. 2025*

- Designed and implemented a geospatial workflow integrating Sentinel-2 analysis and UAV-based instance segmentation (YOLOv8) to automate the detection of Acrostichum aureum with 90.4% precision. This technical innovation successfully bridged macro and micro-scale data to optimize site selection accuracy for the Mangroves for Coastal Resilience (M4CR) project
- DOI: 10.13140/RG.2.2.21992.53762

Assessment of Ecosystem Services and Environmental Degradation by Using Landscape Approach in The Riparian Area of Lake Rawapening

Jurnal GEA

Paper *Oct. 2022*

- This study aims to identify the spatial distribution of ecosystem services provided in part of the riparian area of Lake Rawapening using a geomorphological approach with detailed scale mapping and identify environmental degradation that occurred in the area in two time periods, 2016 and 2020.
- DOI: 10.17509/gea.v22i1.39976

Laporan Akhir Pengelolaan Lingkungan SubDAS Tukbening: Wilayah Cakupan Kab. Semarang dan Kab. Grobogan

Fieldwork 3 Environmental Geography

Report *Oct. 2022*

- Identified environmental challenges in the Tukbening Sub-watershed, such as environmental issues, resources, and management.
- Analyzed appropriate environmental management in the Tukbening Sub-watershed based on land, water, air, and social features.
- Provided direction for sustainable environmental management of land, water, air, and social features in the Tukbening Sub-watershed.
- Link: https://bit.ly/laporankkl3